Shajuisaac Rajendran

Amherst, MA | (413) 230-5565 | shajuisaacrajendran9500@gmail.com | www.linkedin.com/in/shajuisaac

EDUCATION

Master of Science in Mechanical Engineering Aug ‘24 - May ‘26
University of Massachusetts Amherst MA, United States

- Relevant coursework: Thermal Engineering, Advanced Solid mechanics, Additive Manufacturing, Power system designs, Sustainability Engineering for Infrastructures, Technical Program Management, Advanced fluid mechanics, Engineering Leadership and Entrepreneurship

Bachelor of Engineering in Mechanical Engineering Sept ’18 - May ‘22
Anna University Chennai, India

- Relevant coursework: Computer Aided Design, Finite Element Analysis, Strength of Materials, Thermal Engineering, Heat and Mass Transfer, Material Science, Project management

EXPERIENCE

Research And Development Engineer May ’26 - Present
Riccio College of Engineering, UMass Amherst Amherst, MA

- Led prototype mechanical development using CATIA V5/V6 and SolidWorks for CNC machining, fabrication, and rapid prototyping workflows.
- Designed and validated complex 3D mechanical assemblies, sheet metal structures, brackets, mounting interfaces, cable routing.
- Created GD&T manufacturing drawings, BOMs, fabrication callouts, and release documentation for prototype builds.
- Managed cross-functional prototype execution activities across engineering, manufacturing, suppliers, and quality teams during fast-paced development cycles.
- Performed ANSYS structural and thermal validation to identify failure risks, validate thermal performance, and improve hardware reliability before prototype release.
- Supported hands-on prototype builds, fitment validation, assembly troubleshooting, and rapid engineering iterations in lab and manufacturing environments.
- Supported reverse engineering and dimensional validation activities using 3D scanning workflows and inspection methodologies.

Research Assistant Nov ’25 - May ‘26
University of Massachusetts Amherst Amherst, MA

- Developed real time multibody vehicle and tire dynamics models in Dymola using Claytex libraries, enabling Driver in the Loop simulations with 6× faster performance.
- Built and calibrated advanced tire models using FlatTrac and WFT test data, incorporating pressure, wear, and temperature effects to improve vehicle performance correlation.
- Performed vehicle dynamics and suspension analysis including ride & handling, DOE optimization, and NVH studies using Python/C++ workflows.

Design Engineer Intern - R&D Jun ’25 - Aug ‘25
Vectra Automations Inc Cary, NC

- Designed and fabricated custom tools, fixtures, and prototype components using rapid prototyping, CAD/CAE, and DFM methodologies to improve manufacturing productivity and safety.
- Supported prototype manufacturing activities including fabrication, CNC machining coordination, assembly validation, and engineering troubleshooting.
- Conducted dimensional and mechanical root cause analysis on assembly and equipment failures, improving reliability and reducing downtime.
- Designed and released a 1.2 kW motor structural assembly within tight packaging constraints while performing ANSYS structural and thermal simulations across 8 components.
- Automated CAD reconstruction workflows using Python/.NET, reducing early-stage design turnaround time by 30%.
- Generated GD&T compliant supplier manufacturing packages accepted without revision during first submission.
- Maintained technical documentation, issue tracking logs, stakeholder updates, and execution reports across rapid engineering development activities.
- Designed prototype fixtures and supported tooling validation activities to improve assembly repeatability and manufacturing readiness.

Mechanical Design Engineer Sept ’22 - Jun ‘24
Data Patterns India Limited Chennai, India

- Led end-to-end mechanical design and prototype development of transportable radar systems using CATIA V5 and SolidWorks from concept through manufacturing validation and production release.
- Engineered rugged sheet metal, welded structures, and IP67 rated enclosures optimized for fabrication, manufacturability, and harsh environmental operation.
- Developed GD&T compliant fabrication drawings, welding callouts, and detailed manufacturing documentation for CNC machining and sheet metal production.
- Led cross-functional coordination across engineering, manufacturing, suppliers, and quality teams during validation reviews, prototype builds, and production ramp up activities.
- Improved Tier-1 supplier dimensional first-pass yield from 87% to 96% through tolerance stack-up analysis, dimensional variation analysis, and manufacturing process improvements.

- Delivered 12% mass reduction across 12 structural variants through DFM optimization and cost-down initiatives.
- Performed structural strength, buckling, bearing load, and CG optimization analyses to improve system stability and field reliability.
- Applied DFMEA and DVP&R methodologies across mission-critical subassemblies, reducing risk priority numbers by 45%.
- Supported dimensional root cause analysis and assembly fitment troubleshooting for sheet metal tooling and prototype fabrication processes.

Mechanical Engineer Apprentice May ’22 - Aug ‘22
Alkraft Thermotechnologies Chennai, India

- Supported manufacturing operations for thermal management and heat exchanger components.
- Participated in root cause investigations for production-line quality issues.
- Supported preventive maintenance and SOP development to improve line reliability.

PROJECTS

Vehicle Powertrain Alignment & Torsional Dynamics

- Designed, developed and optimized manufacturing processes for EV components including battery packs, motors, and drivetrain systems, conducted time and motion studies to identify inefficiencies increasing throughput.
- Led continuous improvement initiatives using Lean Manufacturing and Six Sigma methodologies to reduce waste and improve process efficiency.

Tail-assisted Stabilization - Bio-inspired Quadruped Locomotion

- Designed 8 structural CAD components for a quadruped tail actuation mechanism using mass distribution, rotational inertia and dynamic load transfer principles, achieved target stabilization moment within 5% of model prediction.
- Evaluated 3 bearing configurations for joint friction minimalization and rolling contact fatigue life, selected configuration reduced estimated joint wear rate by 30% versus baseline.

Analyzing Electric Vehicle Charging & Adoption Trends

- Performed end-to-end data analysis (cleaning, preprocessing, EDA) on EV datasets to uncover adoption trends, vehicle performance patterns, and market growth insights.
- Analyzed temporal, regional, and manufacturer-level trends, identifying key patterns in EV adoption, range distribution, and high-growth segments across U.S. markets.
- Developed data visualizations and applied trend/statistical analysis to communicate insights and support forecasting for infrastructure planning and market demand.

Multi-Body Dynamic Analysis of Quarter Car Suspension Model

- Developed a quarter car suspension model in MSC ADAMS by importing CAD geometry and defining contacts, constraints, forces, and motion inputs for dynamic simulation.
- Simulated various road profiles by tuning spring stiffness and damping coefficients, evaluating system response under different operating conditions.
- Analyzed key vehicle dynamics parameters including wheel travel, slip angle, yaw rate, longitudinal/lateral forces, and tire contact patch behavior using post-processing tools.

Machine Learning-Based Road Condition & Fuel Impact Prediction (Geospatial AI)

- Developed a Python-based geospatial analysis pipeline using A* pathfinding and RSI fuel efficiency modeling to evaluate 6M+ commute pairs, quantifying excess fuel consumption due to road roughness (IRI) and speed constraints.
- Engineered a parallelized data processing framework (concurrent.futures) with checkpoint-based fault recovery, achieving 8× reduction in computation time for large-scale road network analysis.
- Built a data-driven road prioritization model by integrating MassDOT Pavement, Census LODES, and EJ datasets (GeoPandas, OSMnx), classifying 157K+ census blocks and visualizing high-impact segments via interactive Folium geospatial dashboards.

SKILLS

CAD & Modelling: CATIA V5/V6, SolidWorks, Creo Parametric, Siemens NX, AutoCAD, Autodesk Alias, Fusion 360, 3D Assembly Design
Simulation & Analysis: ANSYS Workbench (Structural, Thermal, Modal), FEA, CFD, MSC ADAMS, HyperWorks, Altair MotionView, Thermal Analysis, Tolerance Stack-Up Analysis
Manufacturing & Prototype Engineering: GD&T (ASME Y14.5), DFM, CNC Machining, Sheet Metal Manufacturing, Additive Manufacturing, Prototype Builds, Fabrication, BOM Management, Root Cause Analysis
Programming & Automation: Python, MATLAB, C, VBA, Minitab (Familiar), JMP (Familiar)
PLM & collaboration: JIRA, Confluence, GitHub, MS Excel